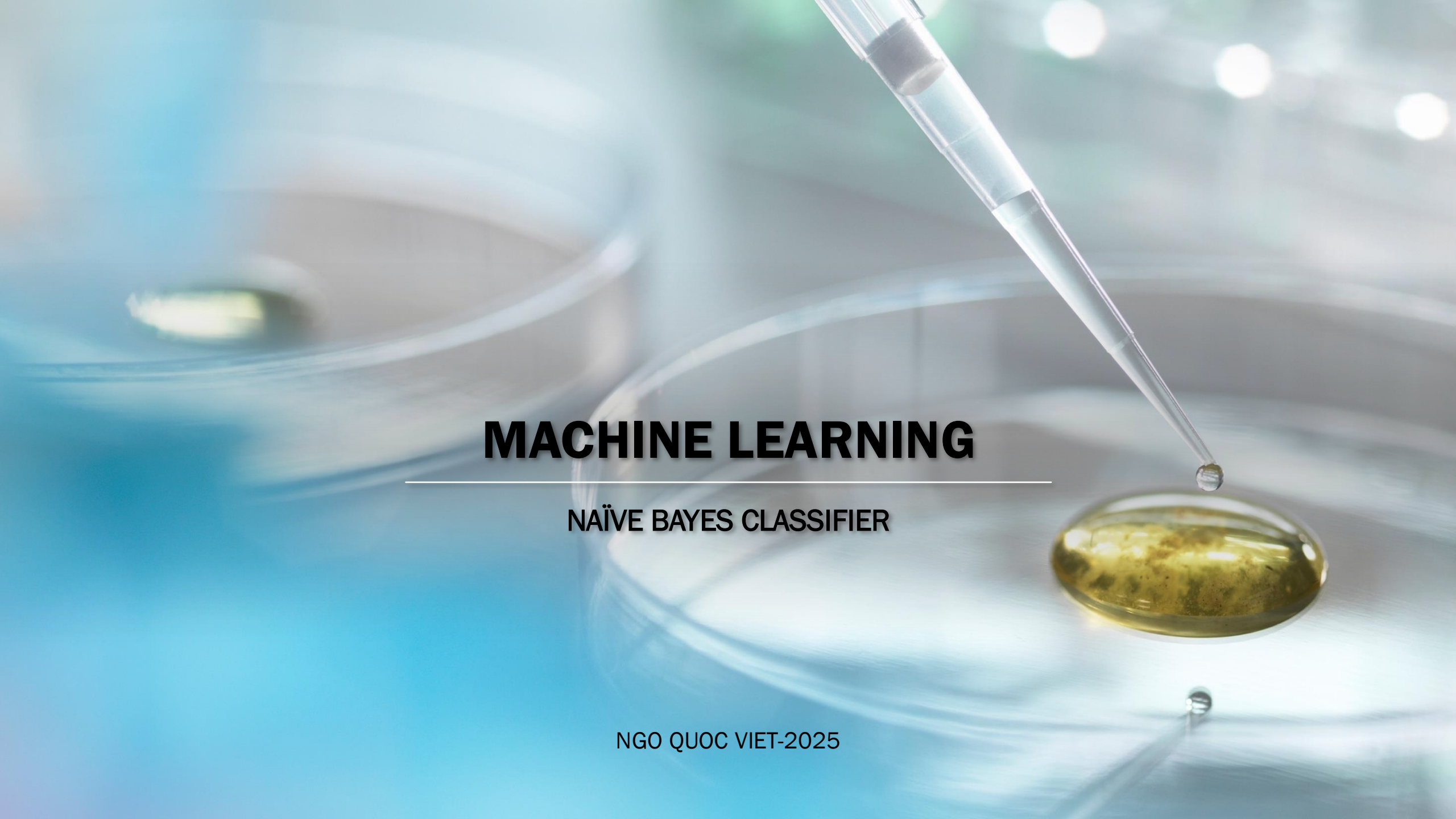


MACHINE LEARNING

NAÏVE BAYES CLASSIFIER

NGO QUOC VIET-2025

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1. Introduction to Naïve Bayes Classifier.
2. The Naive Bayes Classifier with Numerical Attribute Values.
3. Some examples
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Problem statement

- Given features $X_1, X_2, \ldots, X_n$
- Predict a label Y

INTRODUCTION

- Naïve Bayes is a probabilistic machine learning algorithm based on the Bayes Theorem, used in a wide variety of classification tasks.
- It is not a single algorithm but a family of algorithms where all of them share a common principle, i.e. every pair of features being classified is independent of each other. (https://www.geeksforgeeks.org/naive-bayes-classifiers/).
- Naïve?? It is because the classifier assumes that the input features that go into the model are independent of each other.

TYPES OF NAÏVE BAYES CLASSIFIERS



- Optimal Naive Bayes
  - Chooses the class that has the greatest a posteriori probability of occurrence (so-called maximum a posteriori estimation or MAP)
- Multinomial Naïve Bayes Classifier
  - Feature vectors represent the frequencies with which certain events have been generated by a multinomial distribution. This is the event model typically used for document classification
- Bernoulli Naïve Bayes Classifier
  - features are independent booleans (binary variables) describing inputs. Like the multinomial model, this model is popular for document classification tasks, where binary term occurrence features are used rather than term frequencies
- Gaussian Naïve Bayes Classifier
  - Continuous values associated with each feature are assumed to be distributed according to a Gaussian distribution

BAYES THEOREM

Likelihood

$$P(h|D) = \frac{P(D|h).P(h)}{P(D)}$$

Prior.

Calculated by assuming equiprobable classes, ($P(Y) = \frac{1}{K}$), with K is number of classes, or class probability from the training set (*#samples in class / total samples*).

Probability of evidence/data (regardless of the classes). Usually, It is constant..

- P(h) : prior probability of hypothesis h.
- P(D) : prior probability of training data D.
- P(h|D) : conditional probability of h given D.
- P(D|h) : conditional probability of D given h.

We want the most probable hypothesis given the training data

CHOOSING HYPOTHESES

- Generally want the most probable hypothesis given the training data

- **Maximum a posteriori** hypothesis h_{MAP}:

$$h_{MAP} = arg \max_{h \in H} P(h|D) = arg \max_{h \in H} \frac{P(D|h)P(h)}{P(D)}$$

$$= arg \max_{h \in H} P(D|h).P(h)$$

CHOOSING HYPOTHESES

- If assume $P(h_i) = P(h_j)$ for all h_i and h_j ($P(h_i) \approx 1/K$, with K is number of classes), then can further simplify, and choose the:
- **Maximum likelihood** (ML) hypothesis

$$\boldsymbol{h_{MAP} = arg \max_{h_j \in H} P(h_j|D) = arg \max_{h_j \in H} P(D|h_j)}$$

Everything follows two simple rules

Sum rule: $P(x) = \sum_y P(x, y)$

Product rule: $P(x, y) = P(x)P(y|x)$

NAÏVE BAYES CLASSIFICATION EXAMPLE

- Does patient have cancer or not?

A patient takes a lab test and the result comes back positive. The test returns a correct positive result (+) in only 98% of the cases in which the disease is actually present, and a correct negative result (-) in only 97% of the cases in which the disease is not present.

Furthermore, 0.008 of the entire population have this cancer.

It is found that, $(P(cancer) = 0.008) \neq (P(\neg cancer) = 0.992)$, *so we don't have the* assume of $P(h_i) = P(h_j)$ for all h_i and h_j.

NAÏVE BAYES CLASSIFICATION EXAMPLE

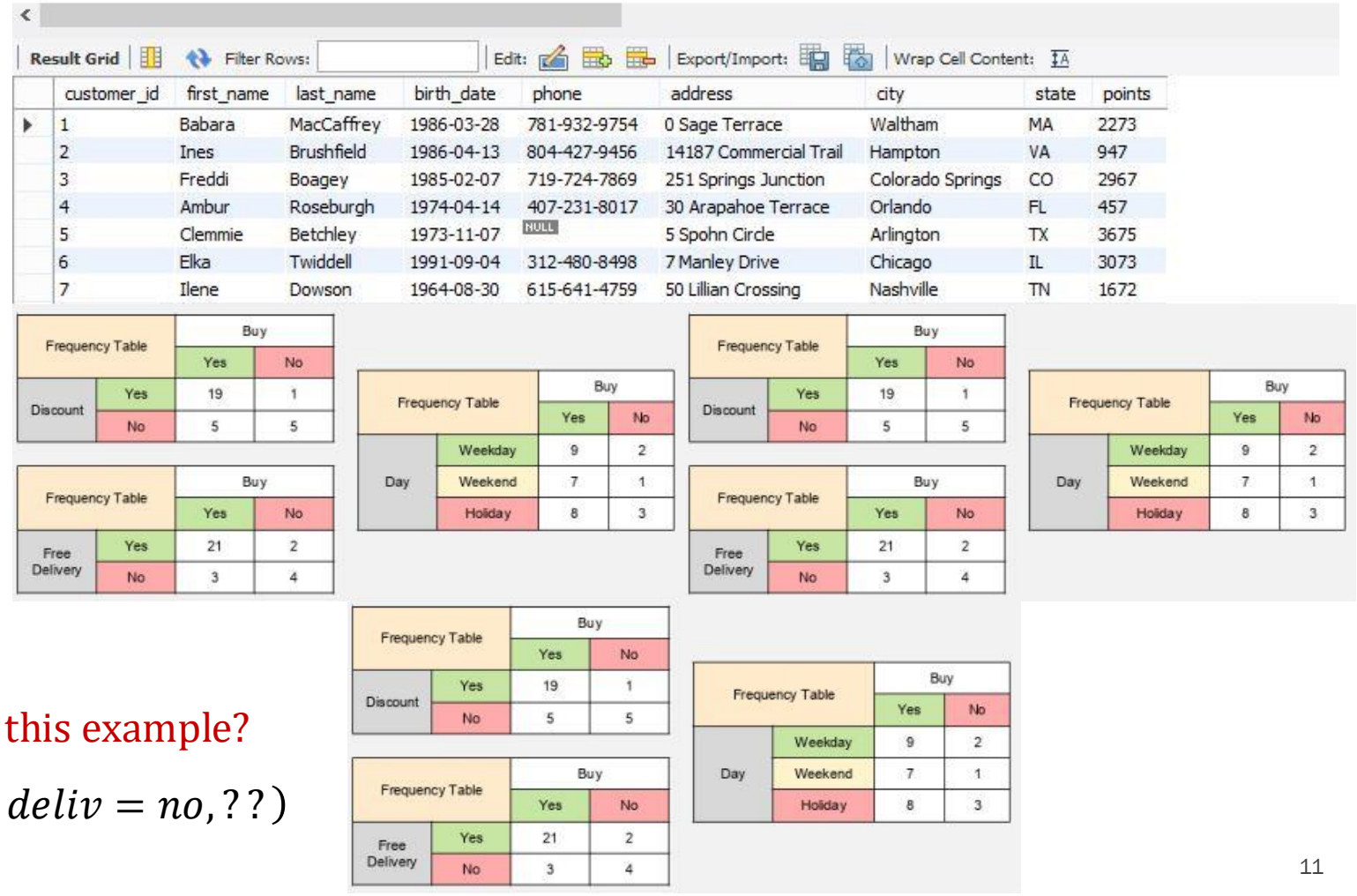
$$P(cancer) = 0.008, \qquad P(\neg cancer) = 0.992$$
$$P(+|cancer) = 0.98, \qquad P(-\,|\,cancer) = 0.02$$
$$P(+|\neg cancer) = 0.03, \qquad P(-\,|\,\neg cancer) = 0.97$$

$$P(cancer|\,+) = P(+|cancer).P(cancer) = 0.98 * 0.008 = 0.0078$$
$$P(\neg cancer|\,+) = P(+|\neg cancer).P(\neg cancer) = 0.03 * 0.992 = 0.0298$$

$$h_{MAP} = \neg cancer$$

NAÏVE BAYES EXAMPLE

- To predict whether a person will purchase a product on a specific combination of day, discount, and free delivery using a Naive Bayes classifier.
- For any given day, check if there are a discount and free delivery. Based on this information, we can predict if a customer would buy the product or not.



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
1	Babara	MacCaffrey	1986-03-28	781-932-9754	0 Sage Terrace	Waltham	MA	2273
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
3	Freddi	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
5	Clemmie	Betchley	1973-11-07	NULL	5 Spohn Circle	Arlington	TX	3675
6	Elka	Twiddell	1991-09-04	312-480-8498	7 Manley Drive	Chicago	IL	3073
7	Ilene	Dowson	1964-08-30	615-641-4759	50 Lillian Crossing	Nashville	TN	1672

Frequency Table		Buy	
		Yes	No
Discount	Yes	19	1
	No	5	5

Frequency Table		Buy	
		Yes	No
Day	Weekday	9	2
	Weekend	7	1
	Holiday	8	3

Frequency Table		Buy	
		Yes	No
Free Delivery	Yes	21	2
	No	3	4

What are h, D in this example?

$$(Holiday, Discount = yes, Free\ deliv = no, ??)$$

NAÏVE BAYES CLASSIFICATION EXAMPLE

Type of family structure	Age group	Income status	Will they buy a car?
Nuclear	Young	Low	Yes
Extended	Old	Low	No
Childless	Middle-aged	Low	No
Childless	Young	Medium	Yes
Single Parent	Middle-aged	Medium	Yes
Childless	Young	Low	No
Nuclear	Old	High	Yes
Nuclear	Middle-aged	Medium	Yes
Extended	Middle-aged	High	Yes
Single Parent	Old	Low	No

$$P(No \mid single\ parent, young, low)$$
$$P(Yes \mid single\ parent, young, low)$$

P(No) = 4/10
P(Yes) = 6/10

P(Single Parent|Yes) = 1/6
P(Single Parent|No) = 1/4
P(Young|Yes) = 2/6
P(Young|No) = 1/4
P(Low|Yes) = 1/6
P(Low|No) = 4/4

P(Yes|X) = P(Single Parent|Yes) * P(Young|Yes) * P(Low|Yes) = 1/6 * 2/6 * 1/6 = 0.0063
P(No|X) = P(Single Parent|No) * P(Young|No) * P(Low|No) = 1/4 * 1/4 * 4/4 = 0.0625

P(Yes|X) = 0.0063/(0.0063 + 0.0625) = 0.09
P(No|X) = 0.0625/(0.0063 + 0.0625) = 0.91

NAÏVE BAYES CLASSIFICATION EXAMPLE

The weather data, with counts and probabilities

outlook			temperature			humidity			windy			play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

A new day

outlook	temperature	humidity	windy	play
sunny	cool	high	true	**?**

NAÏVE BAYES CLASSIFICATION: EXAMPLE

- Likelihood of **yes**

$$= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.0053$$

- Likelihood of **no**

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.0206$$

- Therefore, the prediction is **No.**

THE NAÏVE BAYES CLASSIFIER WITH NUMERICAL ATTRIBUTE VALUES

- One common practice to handle numerical attribute values is to assume **distributions** (Gaussian, Poison, Bernoulli, etc.) for numerical attributes.

The numeric weather data with summary statistics

outlook			temperature		humidity			windy		play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3		83	85		86	85	false	6	2	9	5
overcast	4	0		70	80		96	90	true	3	3		
rainy	3	2		68	65		80	70					
				64	72		65	95					
				69	71		70	91					
				75			80						
				75			70						
				72			90						
				81			75						
sunny	2/9	3/5	mean	73	74.6	mean	79.1	86.2	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	std dev	6.2	7.9	std dev	10.2	9.7	true	3/9	3/5		
rainy	3/9	2/5											

THE NAIVE BAYES CLASSIFIER WITH NUMERICAL ATTRIBUTE VALUES

- Let x_1, x_2, …, x_n be the values of a numerical attribute in the training data set. Gaussian distribution of them is determined

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\sigma = \frac{1}{n}\sum_{i=1}^{n} (x_i - \mu)^2$$

$$f(w) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(w-\mu)^2}{2\sigma^2}}$$

THE NAIVE BAYES CLASSIFIER WITH NUMERICAL ATTRIBUTE VALUES

$$f(\text{temperature} = 66|\text{Yes}) = \frac{1}{\sqrt{2\pi}(6.2)} e^{-\frac{(66-73)^2}{2(6.2)^2}} = 0.0340$$

$$\text{Likehood of YES} = \frac{2}{9} \times 0.0340 \times 0.0221 \times \frac{3}{9} \times \frac{9}{14} = 0.000036$$

$$\text{Likehood of NO} = \frac{3}{5} \times 0.0291 \times 0.038 \times \frac{3}{5} \times \frac{5}{14} = 0.000136$$

THE BAYES CLASSIFIER

- In class, we saw that a good strategy is to predict:

$$\arg\max_{Y} P(Y|X_1, X_2, \cdots, X_n)$$

  - (for example: what is the probability that the image represents a 5 given its pixels?)

- So … how do we compute that?

THE BAYES CLASSIFIER

- Use Bayes Rule!

Likelihood

Prior.
Calculated by assuming equiprobable classes, ($P(Y) = \frac{1}{K}$), with K is number of classes, or class probability from the training set (*#samples* / *total samples*).

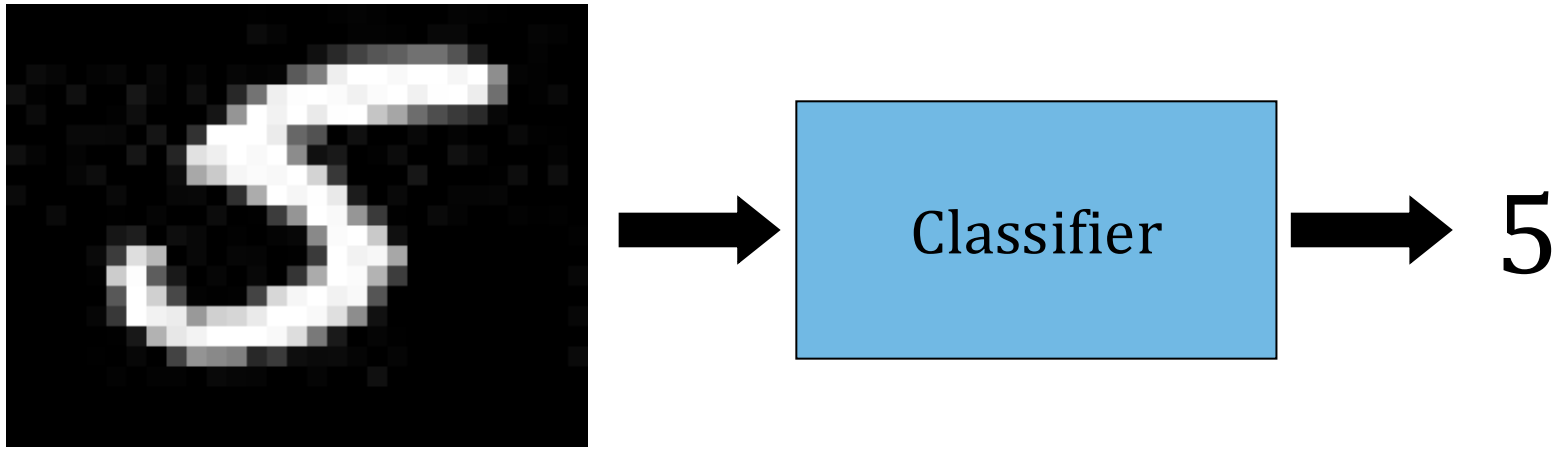
$$P(Y|X_1, X_2, \cdots, X_n) = \frac{P(X_1, X_2, \cdots, X_n|Y).P(Y)}{P(X_1, X_2, \cdots, X_n)}$$

Probability of evidence/data (regardless of the classes). Usually, It is constant.

- Why did this help? Well, we think that we might be able to specify how features are "generated" by the class label.

NAÏVE BAYES CLASSIFICATION EXAMPLE

- Digit Recognition



- $X_1, \ldots, Xn \in \{0,1\}$ (Black vs. White pixels)
- $Y \in \{5,6\}$ (predict whether a digit is a 5 or a 6)

NAÏVE BAYES CLASSIFICATION EXAMPLE

- Let's expand this for our digit recognition task:

$$P(Y = 5|X_1, \dots, X_n) = \frac{P(X_1, \dots, X_n|Y = 5)P(Y = 5)}{P(X_1, \dots, X_n|Y = 5)P(Y = 5) + P(X_1, \dots, X_n|Y = 6)P(Y = 6)}$$

$$P(Y = 6|X_1, \dots, X_n) = \frac{P(X_1, \dots, X_n|Y = 6)P(Y = 6)}{P(X_1, \dots, X_n|Y = 5)P(Y = 5) + P(X_1, \dots, X_n|Y = 6)P(Y = 6)}$$

- To classify, we'll simply compute these two probabilities and predict based on which one is greater

MODEL PARAMETERS FOR DIGIT RECOGNITION EXAMPLE

- For the Bayes classifier, we need to "learn" two functions, the likelihood and the prior.
- How many parameters are required to specify the prior for our digit recognition example? → Just 1 parameter (WHY IN THIS EXAMPLE??)

- How many parameters are required to specify the likelihood?
  - (Supposing that each image is 30x30 pixels)

Too many params

$$2 * (2^{900} - 1)$$

THE NAÏVE BAYES MODEL FOR DIGIT RECOGNITION EXAMPLE

- The *Naïve Bayes Assumption*: Assume that all features are independent **given the class label Y**
- Equationally speaking:

$$P(X_1, \dots, X_n|Y) = \prod_{i=1}^{n} P(X_i|Y)$$

- Number of parameters for modeling for this problem is $\boldsymbol{2n}$.

NAÏVE BAYES LEARNING

Learn $P(X_i|\ Y_j)$; $P(Y_j)$, in which $Y_j \in \{classes\}$, X_i are attributes of each attribute.

- For Naïve Bayes with categorical: creating the frequency tables.
  - For example: in $Y \in \{5,6\}$ binary digit recognition, there are 2 classes, and 900 pixels for each image. With set of trained images, we calculate occurrences/frequencies of black or white at each pixels.
- For Gaussian Classifier with numerical attribute values: learn all tuples of (μ_y, σ_y) of $f(X_i|Y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X_i-\mu_y)^2}{2\sigma_i^2}}$, in which dataset is partitioned by $Y \in \{classes\}$, and attributes.
- We can combine categorical and numeric attributes in Naïve Bayes classification.

BRUTE-FORCE BAYES CONCEPT LEARNING

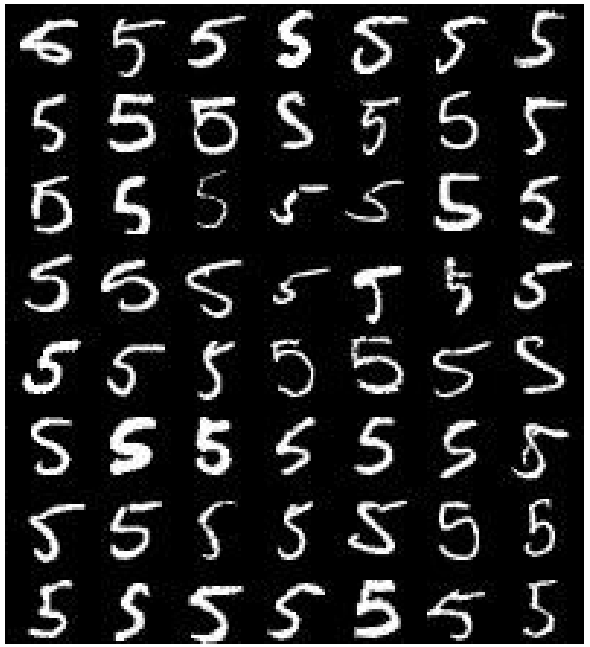
- For each hypothesis h in H, calculate the posterior probability

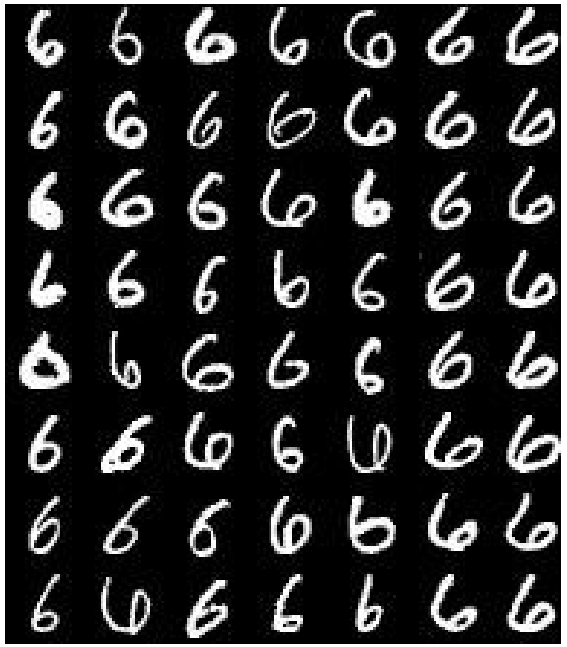
$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

- Output the hypothesis h_{MAP} with the highest posterior probability

NAÏVE BAYES TRAINING

- Now that we've decided to use a Naïve Bayes classifier, we need to train it with some data:





NAÏVE BAYES TRAINING

- Estimate $P(Y=v)$ as the fraction of records with $Y=v$

$$P(Y = v) = \frac{count(Y = v)}{\#records}$$

- Estimate $P(X_i=u|Y=v)$ as the fraction of records with $Y=v$ for which $X_i=u$.

$$P(X_i = u|Y = v) = \frac{Count(X_i = u \wedge Y = v)}{\#records}$$

NAÏVE BAYES TRAINING

- When you have a model with many features, the entire probability will become zero because one of the feature's value was zero. To avoid this, we increase the count of the variable with zero to a small value (usually 1).
- Fix this by adding "virtual" counts:

$$P(X_i = u|Y = v) = \frac{Count(X_i = u \wedge Y = v) + 1}{\#records + 2}$$

SOME COMMENTS OF NAÏVE BAYES CLASSIFIER

- Naïve Bayes is often a good choice if you don't have much training data. Naïve Bayes works best only for small to medium sized data sets.
- Assumption: all features are independent given the class label Y
  - Actually, the Naïve Bayes assumption is almost never true.
  - Still… Naïve Bayes often performs surprisingly well even when its assumptions do not hold.
- GaussianNB works best for continuous types of data.
- Some variants: BernoulliNB, MultinomialNB, ComplementNB.

NAÏVE BAYES CLASSIFIER ADVANTAGES/DISADVANTAGES

Advantages

- Naive Bayes is easy to grasp and works quickly to predict class labels. It also performs well on multi-class prediction
- When the assumption of independence holds, a Naive Bayes classifier performs better compared to other models like logistic regression, and you would also need less training data
- It performs well when the input values are categorical rather than numeric. In the case of numerics, a normal distribution is assumed to compute the probabilities (a bell curve, which is a strong assumption)

Disadvantages

- If a categorical variable has a category in the test data set, which is not observed by the model in the training data, it will assign a 0 (zero) probability and not be able to make a prediction. This is often referred to as "Zero Frequency". To solve this problem, we can use the smoothing technique. One of the simplest smoothing techniques is called Laplace estimation.
- The other limitation of Naive Bayes is the assumption of independence among the features. In real life, it is almost impossible that we get a set of features that are completely independent

LAPLACE SMOOTHING OF THE NB CLASSIFIER

- NB classifier works well enough in text classification problems such as spam filtering and the classification of reviews as positive or negative
- Example of text classification where the task is to classify whether the review is positive or negative. We build a likelihood table based on the training data. While querying a review, we use the Likelihood table values, but what if a word in a review was not present in the training dataset. Example:

$$Query\ review\ =\ w_1 w_2 w_3 w'$$

- We have four words in our query review, and let's assume only w_1, w_2, and w_3 are present in training data. To calculate whether the review is positive or negative, we compare $P(positive|review)$ and $P(negative|review)$.

$$P(positive|review) \propto P(w_1|positive) \times P(w_2|positive) \times P(w_3|positive) \times P(w'|positive) \times P(positive)$$

- $P(w'|review) = 0$, so $P(positive|review) = 0, P(negative|review)$. This is the problem of zero probability.

LAPLACE SMOOTHING OF THE NB CLASSIFIER

- Laplace smoothing is a smoothing technique that handles the problem of zero probability in Naïve Bayes. We can represent

$$P(w' \mid positive) = \frac{number\ of\ reviews\ with\ w'\ and\ y = positive + \alpha}{N + \alpha * K}$$

  - α represents the smoothing parameter,
  - K represents the number of dimensions (features) in the data, and
  - N represents the number of reviews with y=positive
  - If we choose a value of alpha!=0, the probability will no longer be zero even if a word is not present in the training dataset
- Using higher alpha values will push the likelihood towards a value of 0.5, i.e., the probability of a word equal to 0.5 for both the positive and negative reviews. Since we are not getting much information from that, it is not preferable. Therefore, it is preferred to use alpha=1.

NAÏVE BAYES CLASSIFIER SIMPLE CODE

```
weather =
['Sunny','Sunny','Overcast','Rainy','Rainy','Rainy','Overcast','Sunny','Sunny','Rainy','Sunny','Overcast','Overcast','Rainy']

temp =
['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild','Mild','Mild','Hot','Mild']

play =
['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes','Yes','No']

# Import LabelEncoder

from sklearn import preprocessing

# Creating labelEncoder

le = preprocessing.LabelEncoder()

# Converting string labels into numbers.

weather_encoded=le.fit_transform(weather)

temp_encoded = le.fit_transform(temp)

label = le.fit_transform(play)

from sklearn.naive_bayes import GaussianNB

model = GaussianNB()

# Train the model using the training sets

model.fit(com,label)

from sklearn.naive_bayes import GaussianNB

model = GaussianNB()

# Train the model using the training sets

model.fit(com,label)

# Predict Output

predicted= model.predict([[0,2]]) # 0:Overcast, 2:Mild
```

NAÏVE BAYES CLASSIFIER SIMPLE CODE

- https://www.youtube.com/watch?v=O2L2Uv9pdDA
- https://www.datacamp.com/community/tutorials/naive-bayes-scikit-learn
- https://www.geeksforgeeks.org/naive-bayes-classifiers/

THANK YOU